

10. Overview and concepts

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Ensemble Model: the Random Forests

The term "ensemble" means *a group of things or people acting or being taken together as a whole*¹. Unlike the Naive Bayes or SVM algorithm which operates with a single predictor or estimator, an ensemble learner combines the predictions of *hundreds to thousands of estimators* built using a pre-specified algorithm. An ensemble model offers better generalisability and/or robustness than a single estimator.

There are various types of ensemble models designed to solve regression and classification problems. One of the most popular ones is the **Random Forests** model. The model works by building multiple estimators based on the *Decision Tree* algorithm. Random Forests can be used to solve regression and classification.

To comprehend the inner workings of the Random Forests algorithm, it is necessary to gain a basic understanding of the Decision Tree algorithm.

¹ Definition of **ensemble** from the **Cambridge Advanced Learner's Dictionary & Thesaurus** - (<https://dictionary.cambridge.org/dictionary/english/>). © Cambridge University Press

Decision Tree

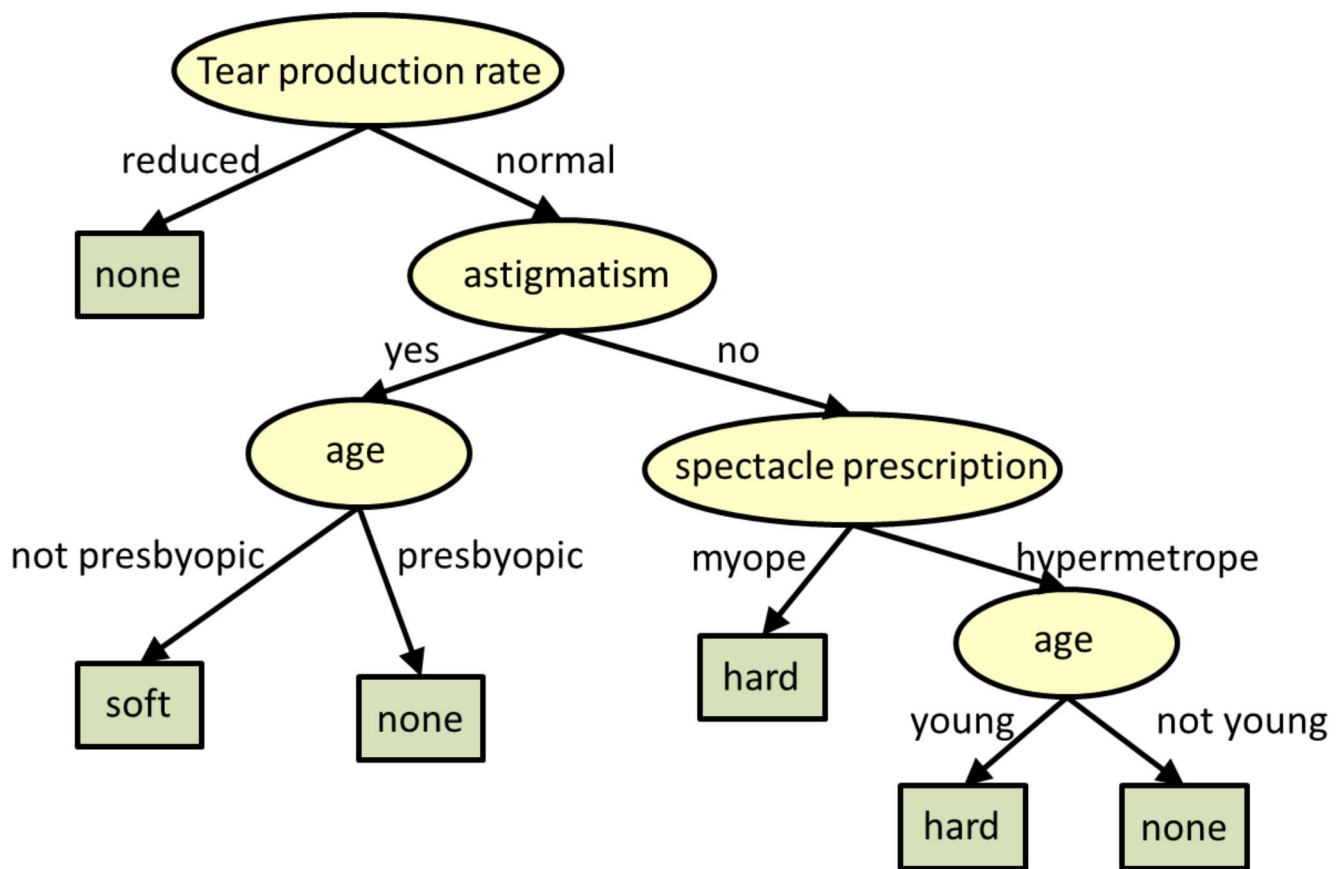
A decision tree is a tree-like model used for classification and regression. In graph theory, a *tree* is a graph with unique characteristics. In the study of discrete mathematics, data structure, and computer science, the topic of *tree* is a vast subject of its own and will not be discussed in detail here. If you are interested to know more about the tree representation, watch the short video below:

Introduction to tree algorithms | Graph Theory



As the name suggests, a decision tree uses a **tree** representation of data to make **decisions**. In data analytics, decisions to be made typically include either (a) assigning a data point to a certain class (*classification*), or (b) assigning a data point with a specific continuous value (*regression*).

Below is an example of a decision tree that determines what kind of contact lens [**soft, hard, none**] a person should wear. The type of contact lens is the *class* (the response variable). The decision tree then assigns the correct contact lens based on a set of explanatory variables (e.g. tear production rate, spectacle prescription, astigmatism, age) about the person.



Source: Carnegie.Mellon.University_(<https://www.cs.cmu.edu/~bhiksha/courses/10-601/decisiontrees/>).

A decision tree is trained from a training set, where the learning algorithm recursively learns to build a tree as follows (as described by Raj 2022; [Carnegie Mellon University](https://www.cs.cmu.edu/~bhiksha/courses/10-601/decisiontrees/) (<https://www.cs.cmu.edu/~bhiksha/courses/10-601/decisiontrees/>)):

- 1) Assign all training instances to the root of the tree. Set the current node to the root node.
- 2) For each attribute:
 - a. Partition all data instances at the node by the value of the attribute.
 - b. Compute the information gain ratio from the partitioning.
- 3) Identify a feature that results in the greatest information gain ratio. Set this feature to be the splitting criterion at the current node. If the best information gain ratio is 0, tag the current node as a leaf and return.
- 4) Partition all instances according to the attribute value of the best feature.
- 5) Denote each partition as a child node of the current node.
- 6) For each child node:
 - a. If the child node is “pure” (has instances from only one class) tag it as a leaf and return.
 - b. If not set the child node as the current node and recurse to step 2.

Random Forests: Overview

Relying on a single Decision Tree may not give the most accurate regression or classification results. Random Forests overcomes this limitation by growing *many* decision trees (a forest). To predict the value of a response variable, the algorithm passes the values of explanatory variables down to each tree in the forest. Each tree then gives its predicted value (a 'vote') and Random Forests take the majority vote (alternatively, the mean of predicted continuous values is taken in the case of regression) as the final predicted value.

Each tree is grown as follows (as described by Breiman and Cutler, 2022; [University of California, Berkeley \(https://www.stat.berkeley.edu/~breiman/RandomForests/cc_home.htm#overview\)](https://www.stat.berkeley.edu/~breiman/RandomForests/cc_home.htm#overview)):

- 1) If the number of instances in the training set is N , sample N cases randomly from the original data - but with replacement. This sample will be the training set for growing the tree.
- 2) If there are M explanatory variables, a number $m < M$ is specified such that at each node, m variables are selected randomly from the M and the best split among these m variables is used to split the node. The value of m is held constant when growing the forest.
- 3) Each tree is grown to the largest extent possible. There is no pruning.

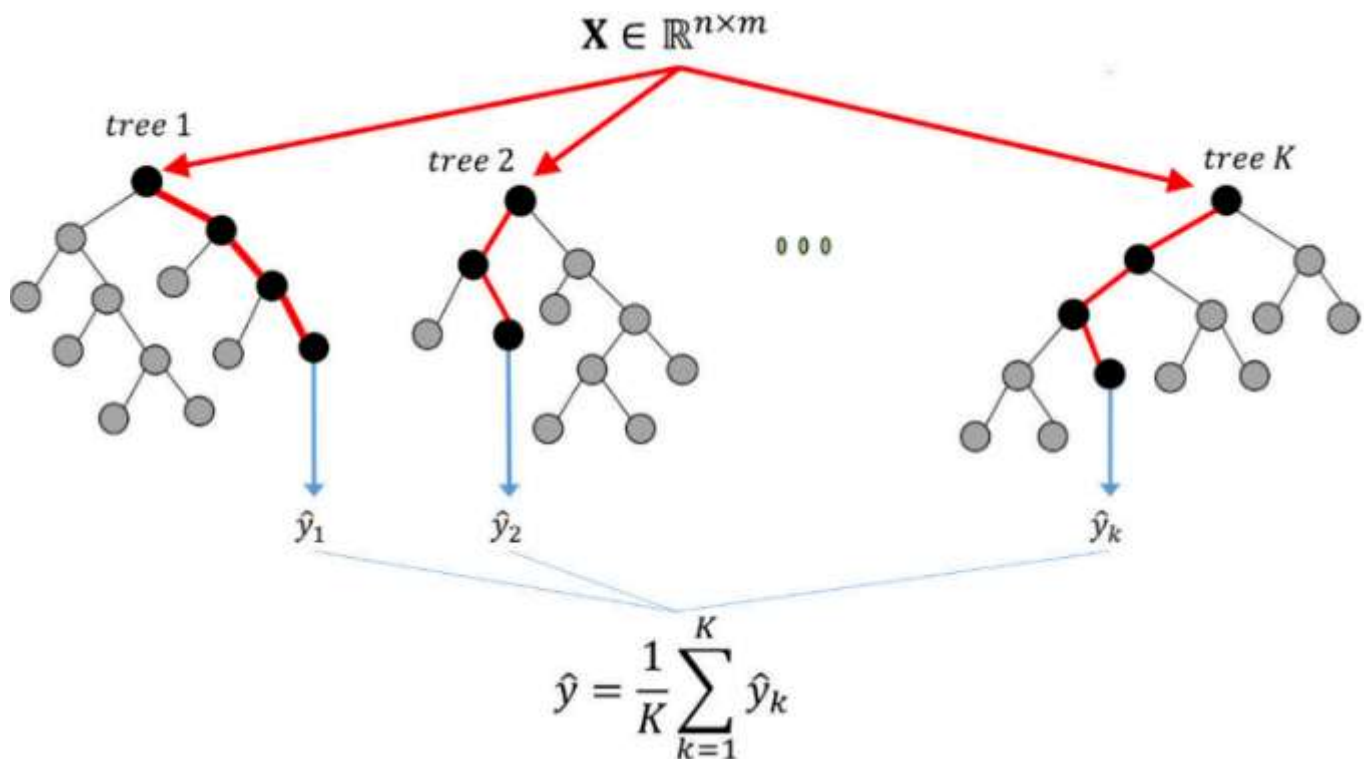
This video visually explains the basic workings of the Random Forests model. Can you explain the purpose of bootstrapping, bagging, and feature randomness?

Visual Guide to Random Forests



Random Forests for Regression

For regression tasks, Random Forests takes the means of all predicted continuous values. The diagram below broadly illustrates the process. In the illustration below, the final prediction $\hat{y}$ is the mean of y values predicted by all trees in the forest.

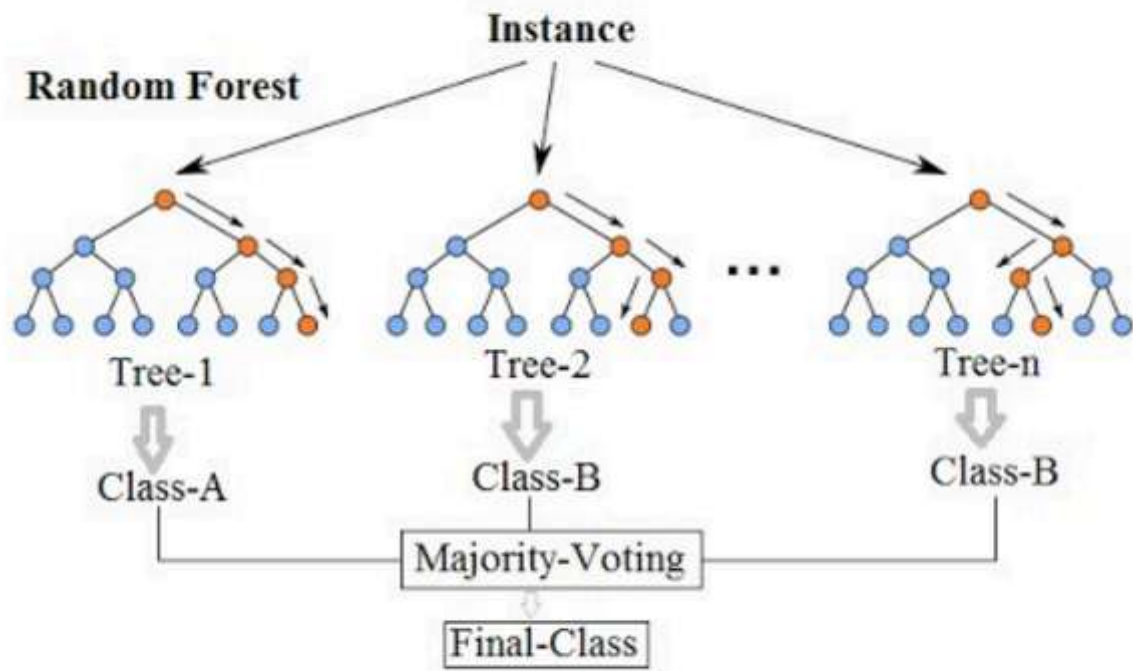


Source: Aldrich.(2020).(<https://www.mdpi.com/2075-163X/10/5/420>).

Random Forests for Classification

For classification tasks, Random Forests work the same way as regression tasks, except that it picks the class label that receives the most votes by individual trees, as illustrated below.

Random Forest Simplified



Source: Wikipedia (https://en.wikipedia.org/wiki/Random_forest)